Akash Arvind Kore

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Summary

AI/ML Engineer with **4+ years** across ML, backend, and cloud engineering, now building production **Generative AI** and **LLM systems** in **Python**: **RAG** pipelines, **Agentic AI** (LangChain, LangGraph, MCP), **multi-agent orchestration**, **LLM evaluation**, and **fine-tuning** (LoRA/QLoRA). Shipped an open-source LLM evaluation framework and a multi-LLM routing layer that cut inference cost **35%** and manual review time **90%** at **99.9% uptime**. M.S. Computer Science, NC State (USA); **3 years** prior engineering experience at **Siemens Digital**, **Principal Global Services**, and **TATA Research**, Pune, India.

Experience

AI Software Engineer Aug. 2025 – Present
Ready Tensor, Inc. *San Diego, CA, USA*

- Developed **readytensor**, an open-source PyPI package for **LLM evaluation and observability** scoring AI/ML repositories on 85 criteria via GPT-4o with **structured outputs** (Instructor) and **prompt engineering**, cutting manual review time by **90%**.
- Architected a **multi-LLM routing layer** spanning OpenAI, Gemini, Groq, Anthropic, and OpenRouter with **function calling**, **tool use**, and **LangChain** orchestration, reducing inference cost by **35%** for a 10-person team.
- Automated production **MLOps pipelines** (Docker, AWS Lambda, GitHub Actions **CI/CD**) at **99.9% uptime**; ran zero-downtime **PostgreSQL** migrations and set **IaC** standards adopted team-wide.
- Delivered **REST APIs** and full **authentication system** (OAuth, RBAC, email verification) with **PII handling** and access governance for a platform migration serving **50,000+ users**.

Machine Learning Intern May 2024 – Aug. 2024
Japan Atomic Energy Agency (JAEA) : National nuclear research lab *Raleigh, NC, USA (Remote)*

- Independently designed a **computer vision** pipeline for nuclear-reactor bubble detection, selecting **Pix2Pix GAN** (PyTorch) for sim-to-real **domain adaptation** under limited labeled data; reached **90% detection accuracy** and **30% better generalization**.
- Curated a 50k-image labeled dataset and applied **transfer-learning** fine-tuning; containerized the **real-time inference pipeline** (Docker, AWS Lambda) at **15 FPS** with **95% lower overhead**.

Software Engineer Associate Sep. 2021 – Jun. 2023
Siemens Digital Industries Software *Pune, India*

- Engineered Teamcenter on Cloud (TcX) on **AWS (EC2, S3, RDS, Lambda)** for **500+ global enterprise customers**; automated **CI/CD** with CloudFormation, cutting deployment setup time by **30%**.
- Resolved **40+ critical production incidents** in Python, C++, and Bash across **Agile sprints**; refactored **C++ server-side microservices**, cutting bug-resolution time by **25%**.

Software Engineer Intern Jan. 2021 – Sep. 2021
Principal Global Services *Pune, India*

- Built Retirement Income Forecaster (**Django REST** + React) at sub-200ms API latency; automated AWS Lambda **data pipelines**, cutting report time **40%**.

Research Intern May 2020 – Jul. 2020
Tata Research Development and Design Centre (TRDDC) *Pune, India*

- Built graph-based **anomaly detection** library (OddBall, Scikit-learn) improving accuracy **15%**; shipped as **PyPI package** used by **5+ hospital partners** at **98% reliability**.

Projects

B.I.A.S.E.D. : Agentic RAG Decision Intelligence Platform [GitHub] [Case Study] *Nov. 2024 – Mar. 2025*
FastAPI, Next.js, pgvector, LangGraph, LangChain, MCP, Ollama, Docker | Apache-2.0

- Implemented an **MCP server** with **function calling** and **tool use** inside a **LangGraph multi-agent** layer routing queries to the best model, returning cited, schema-validated responses grounded in **pgvector** RAG.
- Released full open-source stack (Next.js PWA + FastAPI, **RBAC** auth, Docker Compose) with a scenario-planning engine for multi-hypothesis decision support; Apache-2.0 licensed.

Receipt Intelligence Pipeline [GitHub] [Case Study] *Apr. 2025 – May 2025*
AWS Textract, GPT-4o, Instructor, Celery, FastAPI, PostgreSQL, Redis, Docker

- Production **OCR + LLM** pipeline with **Celery** async workers and typed **Pydantic** extraction; low-confidence outputs routed to **human-in-the-loop** review and fed back as **weak supervision** for retraining; **90%** CORD v2 benchmark success.

Technical Skills

Languages: Python, SQL, TypeScript, JavaScript, C++, Bash

Agentic AI & Orchestration: LangChain, LangGraph, CrewAI, AutoGen, MCP, Function Calling, Multi-Agent Systems, Agentic Workflows

Generative AI & LLMs: GPT-4o, Claude, Gemini, LLaMA-3, OpenAI API, RAG, Prompt Engineering, Fine-Tuning (LoRA/QLoRA), PEFT, LLM Evaluation, LangSmith, AI Observability, Human-in-the-Loop

Data & Vector DBs: Apache Spark, Airflow, Kafka, ETL Pipelines, PostgreSQL, MongoDB, Redis, pgvector, Pinecone, ChromaDB, FAISS, Embeddings

MLOps & Cloud: AWS (S3, Lambda, SageMaker, Bedrock), Azure, Azure OpenAI, GCP, Docker, Kubernetes, CI/CD, MLflow

AI / ML: PyTorch, TensorFlow, Hugging Face, Scikit-Learn, NLP, Computer Vision, Model Deployment

Backend & APIs: FastAPI, Django, REST APIs, Microservices, Pydantic, Next.js, React.js, Git

Education

M.S., Computer Science *North Carolina State University, Raleigh, NC, USA* Aug. 2023 – May 2025

B.Tech, Computer Science *Walchand College of Engineering, Sangli* Aug. 2017 – Jul. 2021